

Belal Ahmed Mohammed Hamed

Minia , Minia , Egypt

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Date of Birth: Sep 25, 1997

Place of Birth: Minya, Egypt

Citizenship: Egyptian

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Education

Minya University, Minya-Egypt, PhD (Computer Science) (2023 – 2026)

- Study emphases: Enhancing Disease Diagnosis with AI-Powered Genomic Analysis.
- PhD's thesis: "Enhancing Disease Diagnosis with AI-Powered Genomic Analysis " – Advisors: Prof. Dr. Tarek Abd El-Hafeez, Dr. Ahmed Omer and Dr. Heba Mamdouh.

Minya University, Minya-Egypt, Master (Computer Science) (2020 – 2023)

- Study emphases: Pattern Matching Algorithms in DNA Sequences.
- Master's thesis: "Improving Pattern Matching Algorithms for Biological Sequences " – Advisors: Prof. Dr. Tarek Abd El-Hafeez and Dr. Osman Ali.

Minya University, Minya-Egypt, Bachelor (Computer Science) (2015 – 2019)

- Study emphases: Computer Systems, Computer Languages, Databases, Algorithms, Information Systems, Object Oriented Programming, Data Structures, Assembly, Operations Research, System Analysis and Design, Operating Systems, Computer Graphics, Visual Programming, Files Organization, Formal Languages, Software Engineering, Computer Graphic Theory, Information Security, Logic Programming (Prolog), Artificial Intelligence (AI), Compiler, Image Processing, Genetic Algorithms, Simulation and Modeling, Computer Networks, Computer Vision, Natural Language Processing.
- Final Grade: 3.45 from 4 GPA Points.

Tell Bani Omran Secondry School, Minya-Egypt, High School (2012 – 2015)

- Final Grade: 91.7% (375.5 from 410).

Publications

1. Ibrahim, O.A.S., Hamed, B.A. & El-Hafeez, T.A. A new fast technique for pattern matching in biological sequences. J Supercomput (2022). <https://doi.org/10.1007/s11227-022-04673-3>
2. Hamed, Belal A., Osman Ali Sadek Ibrahim, and Tarek Abd El-Hafeez. "A Survey on Improving Pattern Matching Algorithms for Biological Sequences." . <https://doi.org/10.1002/cpe.7292>

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3. Hamed, B.A., Ibrahim, O.A.S. & Abd El-Hafeez, T. Optimizing classification efficiency with machine learning techniques for pattern matching. J Big Data 10, 124 (2023). <https://doi.org/10.1186/s40537-023-00804-6>.
4. Hamed, B.A., Farghaly, H.M., Omar, A. et al. Identifying key genetic variants in Alzheimer's disease progression using Graph Convolutional Networks (GCN) and biological impact analysis. J Big Data 12, 171 (2025). <https://doi.org/10.1186/s40537-025-01228-0>
5. Hamed, B. A., Farghaly, H. M., Omar, A., & Abd El-Hafeez, T. A Novel Deep Learning Framework for Alzheimer's Gene Prediction via SNP and Gene Network Integration. [10.21608/SJDFS.2025.444026.1273](https://doi.org/10.21608/SJDFS.2025.444026.1273)

Academic Honors & Scientific Roles

Journal Reviewer | Springer Nature & Diverse Publishers — Nov 2024 – Present

- Actively serving as a peer reviewer for several high-impact journals, ensuring the quality and scientific integrity of published research in AI, bioinformatics, and IoT.
- **Verified Reviewer for:**
 - *Scientific Reports* (Springer Nature)
 - *Discover Internet of Things* (Springer Nature)
 - *The Journal of Supercomputing* (Springer Nature)
 - *BMC Bioinformatics* (Springer Nature)

Young Scientist Member | Medical Data Mining Journal (TMR Journals) — June 2025 – Present

Recognized as an active supporter and contributor. Eligible for reviewer certification and potential editorial board membership.

Experience

Minya University, Minya-Egypt (Lecturer)

Jan-2026 – now

Preparing and teaching lectures to undergraduate students, supervising graduation projects, and supervising postgraduate students.

Microsoft Data Engineer, DEPI-Egypt (Instructor)

Apr-2024 – Oct 2024

Preparing and giving lab sessions to undergraduate students. And equips students with essential skills in data engineering, cloud technologies, and machine learning.

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Minya University, Minya-Egypt (Assistant Lecturer)

July-2023 – now

Preparing and giving lab sessions to undergraduate students.

Minya University, Minya-Egypt (Teaching Assistant)

Oct-2019 – Jun 2023

Preparing and giving lab sessions to undergraduate students.

ITI, Minya -Egypt (Open Source Developing Trainee)

Jun-2019 – Oct-2019

Acts as individual contributor in team work about developing Open-Source application.

Udacity, Online Training (Android Development Trainee)

Apr-2018 – Jul-2018

About solo work for developing android mobile application.

Leadership Experience

Minya University, Teaching and Supervision fellow: (2025 – now)

- Teaching 5 Minya undergraduate courses ranging in size from 20 - 200 students. Topics included: basic programming, object-oriented programming, Mobile Application, information technology, and Natural Language Processing.
- Preparing course material including lectures, final exams, assignments, and oral exams.
- Supervising students in final graduation projects.
- Supervising postgraduate students.

Minya University, Teaching fellow: (2019 – 2025)

- Assisted in teaching 10 Minya undergraduate courses ranging in size from 20 - 500 students. Topics included: programming languages, database, data structure, object-oriented programming, software engineering, information technology, operating system, artificial intelligence, logic programming (prolog) and computer graphics.
- Prepared course material including laboratory practices, practical exams, homework, and practice problems.
- Led weekly laboratory and/or problem-solving and discussion sections for groups of 20 - 35 students.
- Supervised students in final projects, graded exams, and weekly homework.

ITI Open-Source: Skills include (2019)

- Being creative, hard worker, self-motivated, punctual.

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- Being a good team player and having the ability to work under pressure.
- Having excellent presentation skills.

IT Skills

- **Programming Languages:** MATLAB, Python, C++, C#, Java, Assembly.
- **Programming Paradigms:** Object Oriented Programming.
- **Database Development Tools & Languages:** ADO.NET, Entity Framework, Microsoft SQL Server Management Studio, Microsoft Access, MySQL, SQL Lite, and Oracle.
- **Operating Systems:** All windows versions, Linux OS.
- **Developing Applications:** Microsoft Visual Studio, Microsoft Visual Code, C++, NetBeans, Anaconda , PyCharm, MATLAB, and Android Studio.
- **Mobile SDKs:** Android.

Selected Projects

- **Expenses Manager Application (2022): (Individual Contributor – teamwork)** An application to manage monthly expenses and revenues for all currencies and users.
- **Mondabet Application (2019): (Individual Contributor – teamwork)** A disciplined application to supervise the supervisory tours of some clients in the Kingdom of Saudi Arabia.
- **W3eny Project (2019): (Individual Contributor – teamwork)** A mobile app to reduce the problem of nutritional awareness and help type 2 diabetics get rid of this disease and maintain their healthy food.

Awards

- **(2023 – 2026):** Full Tuition PhD Scholarship for Outstanding Students, Faculty of Science, Minya University.
- **(2020 – 2023):** Full Tuition MSc Scholarship for Outstanding Students, Faculty of Science, Minya University.
- **June 2019:** Very good with Degree of Honor (GPA 3.45), 1st place out of 70 students, Faculty of Science, Minya University, Egypt.
- **(2017 – 2019):** Excellence Students Award, Faculty of Science, Minya University, Egypt.

Languages

- **Arabic (native language).**
- **English (Professional working proficiency).**

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References (Academic & Professionals)

Tarek Abd El-Hafeez, Professor

Computer Science Department

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Abdelmgeid Amin, Professor

Computer Science Department

Minia University, Minia, Egypt

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Ahmed Omer Abd-Allah, Assistant Professor

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Dr. Heba Mamdouh, Assistant Professor

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